

Movie Recommender

Reinforcement Learning Project Outline

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Goal:

- We want to be able to create a recommendation system that can provide accurate movie suggestions after getting feedback on initial movies watched
- We believe that based on movie genres and similarities, we can predict what kind of movie a user may like early on in their interaction with our platform
- Will use part of the data as previous knowledge and new data to simulate a user watching a specific movie we may recommend
- We can implement two scenarios for new users:
 - Knowing previous movies that a new user has watched and recommending based on that
 - Knowing **no** movies that a new user has watched and recommending based on learned popularity of movies

Data: <https://grouplens.org/datasets/movielens/>

- This data is composed of about 32 million movie ratings with at least 20 from each unique user
- Additionally there is a data set that contains synthetic data generated through fractal expansions to include 1 billion observations
- We do not have demographic data, only unique ID for each user, which makes it more difficult initially to recommend and more similar to real world scenarios when gaining a new customer and having no prior information about their preferences